

Computational Image Analysis to Support Microplastic Mitigation in Wastewater Treatment Systems

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Abstract. Microplastics are a widespread environmental pollutant with potential impacts on ecosystems and human health. Although wastewater treatment plants (WWTPs) reduce their release, significant quantities still reach aquatic environments. Traditional detection methods, such as micro-FTIR and Raman spectroscopy, offer high accuracy but are limited in terms of cost, time, and scalability.

Recent advances in computer vision and deep learning enable automated and scalable detection of microplastics from microscopy images. This paper reviews current approaches, including convolutional neural networks, object detection frameworks, and hybrid methods, highlighting key challenges such as limited datasets and poor generalization to real-world conditions.

To address these challenges, a modular methodology based on artificial intelligence is proposed, integrating preprocessing, U-Net-based segmentation, and data augmentation strategies. The system is designed to support scalable and real-time monitoring in wastewater treatment environments.

Keywords: Microplastics · Deep Learning · Computer Vision · Wastewater Treatment

1 Introduction

Microplastics, defined as plastic particles smaller than 5 mm [14], have emerged as a critical environmental concern due to their persistence, ubiquity, and potential impact on ecosystems and human health [1, 2, 14]. They result from the degradation of larger plastic materials as well as from primary sources such as cosmetic products and synthetic fibers.

Recent studies have demonstrated the widespread presence of microplastics in aquatic environments, including freshwater and marine systems [15, 14]. This presence becomes even more evident due to their accumulation in aquatic organisms, such as fish, bivalves, and plankton, indicating successive transfer along the food chain [15, 14].

Furthermore, traces of microplastics have also been detected in human biological samples, including blood, raising concerns about potential health risks [16].

In this context, wastewater treatment plants (WWTPs) play a crucial role in mitigating pollution caused by these particles, removing a significant fraction from aquatic systems [17, 18]. However, current systems are not fully efficient, and substantial amounts of microplastics continue to be released into the environment.

Traditional analytical techniques, such as micro-FTIR and Raman spectroscopy, provide high chemical specificity and remain the gold standard [1, 2]. Despite their accuracy, they are costly, time-consuming, and require specialized expertise.

In this context, computer vision and deep learning offer a promising alternative, enabling automated detection, cost reduction, and large-scale monitoring [3–5].

2 State of the Art

The detection and monitoring of microplastics have evolved significantly, transitioning from traditional spectroscopic techniques to advanced methodologies based on image analysis and machine learning [19]. This evolution reflects the growing need for scalable, efficient, and automated solutions capable of handling the complexity of wastewater matrices and other environmental samples.

2.1 Traditional Spectroscopic Techniques

Spectroscopic approaches, including micro-FTIR and Raman spectroscopy, remain the gold standard for microplastic identification due to their high chemical specificity and accurate polymer characterization capabilities [1, 2]. These methods allow reliable differentiation between polymer types, which is essential for environmental impact assessment.

However, despite their high accuracy (Table 1), these techniques present significant limitations. High operational costs, low throughput, and reliance on expert knowledge hinder their large-scale application [1, 2]. Additionally, sample preparation and analysis times make them unsuitable for real-time monitoring, especially in dynamic environments such as WWTPs [17, 18].

2.2 AI-Assisted Microscopy

To overcome the limitations of spectroscopic techniques, classical image processing methods were initially explored. These approaches rely on manually defined features such as color, morphology, and texture to identify microplastic particles [19]. Although computationally efficient, they exhibit only moderate accuracy and low robustness in heterogeneous samples with variable illumination and high noise [19, 17, 18].

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), represented a significant advancement. These methods enable

automatic feature extraction and improved generalization, achieving high detection accuracy under diverse imaging conditions [5–7]. Compared to classical approaches, they offer a better balance between accuracy and scalability, although they strongly depend on the availability of high-quality annotated datasets [8, 13].

2.3 Object Detection and Segmentation

Within deep learning, two main approaches can be considered, each suited to different analytical needs.

First, object detection approaches such as YOLO prioritize real-time performance and computational efficiency [12]. These are particularly suitable for portable or embedded systems, enabling rapid sample analysis. However, they provide only moderate precision in the exact delineation of particles [12].

On the other hand, segmentation-based approaches, such as Mask R-CNN and U-Net, perform pixel-level classification, allowing precise delineation of microplastics [9, 6]. They enable accurate quantification and detailed morphological analysis, albeit with higher computational cost and lower real-time capability [9, 6]. This highlights a clear trade-off between speed and precision [9, 12].

2.4 Hybrid Approaches

Hybrid approaches combine different detection modalities to leverage their complementary advantages. For instance, integrating spectral imaging with deep learning enables both morphological and chemical characterization of microplastics [10]. Similarly, the use of photoacoustic imaging with automated preprocessing has shown improvements in detection within complex samples [11].

These approaches are particularly advantageous when visual information alone is insufficient, such as in distinguishing visually similar polymers or detecting particles in complex matrices [10, 11]. However, they introduce additional complexity, requiring careful calibration and specialized hardware [10, 11].

2.5 Datasets and Data Augmentation

The performance of deep learning models strongly depends on the availability of representative and well-annotated datasets. Recent works have introduced microscopy datasets for microplastic analysis, contributing to benchmarking and validation [8].

Nevertheless, significant challenges remain, including small dataset sizes, limited diversity, and acquisition under controlled conditions [8]. These issues hinder generalization to real-world data, which is inherently more variable [8, 17, 18].

To mitigate these limitations, data augmentation techniques are employed, including geometric transformations and Generative Adversarial Networks (GANs) [13]. While these strategies improve model robustness, they do not eliminate the need for real and standardized datasets.

2.6 Synthesis and Comparative Evaluation

Microplastics in wastewater systems exhibit complex dynamics due to treatment processes [17, 18]. This variability directly affects the performance of detection methods.

Table 1 summarizes the main approaches studied. It is evident that none of the existing solutions simultaneously satisfies the requirements of accuracy, scalability, and real-time processing, reinforcing the need for integrated systems.

- Traditional spectroscopic methods offer very high accuracy but are limited by low scalability and high cost [1, 2].
- Classical image processing provides moderate performance but lacks robustness in complex environments [19].
- Deep learning approaches achieve a balance between accuracy and automation, though they depend heavily on data availability [5–8].
- Real-time detection models prioritize scalability but sacrifice precision [12].
- Hybrid methods deliver superior accuracy but introduce significant system complexity [10, 11].

Overall, no single approach fully satisfies the requirements of accuracy, scalability, and real-time capability. This reinforces the need for integrated and modular systems capable of balancing these trade-offs in practical deployment scenarios.

Table 1. Comparison of microplastic detection approaches

Method	Accuracy	Automation	Scalability	Limitations
Micro-FTIR [1]	VH	L	L	Expensive, slow, expert-dependent
Raman Spectroscopy [2]	VH	L	L	Costly, sensitive to interference
Classical Image Processing	M	M	H	Low robustness in complex samples
CNN-based Detection [5, 6]	H	H	MH	Requires labeled datasets
Hybrid (DL+Spectral) [10]	VH	M	M	Complex integration and validation
Real-time Detection (YOLO) [12]	M	H	H	Limited validation in diverse conditions

VH = Very High, H = High, M = Medium, L = Low, MH = Medium-High.

3 Research Gap

Despite significant advances in AI-based microplastic detection, several limitations persist. These are often associated with the fact that most deep learning

models are trained on datasets acquired under controlled laboratory conditions [8], which do not adequately represent the variability of real wastewater environments [17, 18]. As a result, performance degrades when these models are applied to real-world data.

Moreover, the lack of large, standardized, and publicly available datasets hinders reproducibility and comparison across different methods [8]. Evaluation protocols are also inconsistent, complicating benchmarking efforts.

Another identified limitation is the fragmentation of existing solutions. Many studies focus on isolated tasks, such as detection or segmentation, rather than developing complete end-to-end systems.

Finally, although hybrid approaches are promising, their integration into scalable systems remains challenging due to complexity and computational requirements [10, 11].

These gaps highlight the need for robust, modular, and scalable systems.

4 Proposed Methodology

To address the identified limitations, this work proposes a modular AI-assisted system for microplastic detection in wastewater environments, integrating image acquisition, preprocessing, segmentation, and classification into a unified pipeline.

The system begins with image acquisition using microscopy techniques adapted to wastewater samples. Preprocessing steps, including noise reduction, contrast enhancement, and normalization, are applied to improve image quality and reduce variability.

For segmentation, a U-Net-based architecture is employed due to its demonstrated effectiveness in biomedical image analysis and microplastic detection tasks. This model enables precise pixel-level identification of microplastic particles, even in complex backgrounds.

To improve robustness and generalization, data augmentation strategies are incorporated, including geometric transformations and synthetic data generation. These techniques address the scarcity of annotated datasets and enhance model performance in diverse environmental conditions.

Feature extraction and classification modules are integrated to enable further characterization of detected particles, supporting downstream analysis and quantification.

The proposed system is designed with scalability and real-time processing capabilities in mind, allowing integration with portable microscopy setups and potential deployment in wastewater treatment facilities. Additionally, the modular architecture facilitates the incorporation of hybrid sensing approaches, enabling future extensions that combine visual and spectral data.

By directly addressing the limitations identified in the state of the art—namely generalization, dataset scarcity, and system integration—this methodology aims to provide a robust and practical solution for large-scale microplastic monitoring.

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